

Network Anomaly Detection for Intrusion Detection Systems Using Q-Learning and Deep Q-Learning: A Reinforcement Learning Approach

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Abstract—Network anomaly detection is very important for making modern cybersecurity systems stronger, especially since cyber threats like zero-day attacks are becoming more complicated. This study looks into how to use Reinforcement Learning (RL), specifically Q-Learning and Deep Q-Learning (DQN), to create an adaptive intrusion detection system that can find both known and new attack patterns. The UNSW-NB15 dataset is used in the study, and a zero-day simulation is created by leaving out the "Fuzzers" and "Reconnaissance" attack types from the training data. Data preprocessing includes feature selection, encoding, normalization, and class balancing with SMOTE-Tomek to address imbalanced data. We compare the two algorithms using different hyperparameter tuning methods. The performance shows that DQN performs better than Q-Learning in all the metrics considered for evaluation. It records a test accuracy of 99.09% and F1-score of 0.9918 in the scenario of normal data. It also holds up well under the presence of zero-day attacks, where the accuracy drops only marginally (0.07%). The findings show that neural-based RL models are better at learning abstract representations of attacks.

Keywords—anomaly detection, reinforcement learning, Q-learning, deep Q-learning, intrusion detection system, zero-day attacks, network security

I. INTRODUCTION

Network systems are of critical importance to how business processes function today but also increasingly vulnerable to advanced cyberattacks like Distributed Denial of Service (DDoS), malware, and zero-day attacks [1]. Signature-based or rule-based intrusion detection systems are not necessarily effective against new attacks and have numerous false positives, which hampers response times and can have extreme consequences [2].

Anomaly-based IDS is also an option because it looks for deviations in normal network traffic. But such systems struggle with adaptation. Normal changes in traffic can result in false alarms, and new ways of attacks could go undetected because there is not enough historical data [3]. These are problems that show how critical it is to have smart, adaptable solutions that learn based on changing network patterns.

Reinforcement Learning (RL) is very promising as it enables agents to learn best ways to find things by interacting with their world. RL models are based on rewards and can also learn in real time to adjust to new situations. They are great for use in

cybersecurity [4]. This study examines and compares DQN and Q-Learning for network anomaly identification using the UNSW-NB15 dataset. For the purpose of evaluating zero-day detection efficacy, two types of attacks, Fuzzers and Reconnaissance, are omitted from training and set aside for testing.

II. RELATED WORKS

One notable piece of work by Modirrousta et al. suggested a Deep Reinforcement Learning (DRL) method based on a Convolutional Neural Network (CNN) structure. UNSW-NB15 and BoT-IoT datasets were used to test their method, which possessed a very high rate of detection accuracy of 99.17% for the DQN and DDQN approaches [5].

Alavizadeh et al. designed a Deep Q-Learning (DQL)-based intrusion detection model in another research study. In the method, NSL-KDD dataset was used to test the integration of Q-Learning with a deep feed-forward neural network. Depending on the type of attack, the model was able to correctly classify 74% to 78% of the cases [4].

Tellache et al. [6] also utilized multi-agent reinforcement learning architecture based on Deep Q-Networks (RL-DQN) for intrusion detection. Their method performed very well with the CIC-IDS-2017 data, achieving accuracies of 99% with a remarkably low false positive rate of 0.16%. Table I shows the comparison results of the RL-based IDS.

TABLE I. COMPARISON OF RL-BASED INTRUSION DETECTION STUDIES

Study	Methodology	Dataset	Accuracy Results	Year
Modirrousta et al. [5]	DRL with CNN	UNSW-NB15, BoT-IoT	99.17%	2022
Alavizadeh et al. [4]	Deep Q-Learning	NSL-KDD	78%	2022
Tellache et al. [6]	Multi-agent RL-DQN	CIC-IDS-2017	99%	2024

These studies demonstrate that RL, specifically Deep Q-Learning, can be well-suited to uncover intrusions. Nonetheless, some research issues remain behind, such as how to uncover zero-day attacks, how to deal with unbalanced data, and how to make models work better in real-world settings.

III. METHODOLOGY

A. Data Preparation

The UNSW-NB15 dataset from the Australian Centre for Cyber Security [7] contains network traffic data with both normal activities and cyberattacks. It includes 45 features (IP addresses, port numbers, protocol types, traffic metrics) and consists of 175,341 training records and 82,332 testing records [7]. To assess zero-day detection capabilities, "Fuzzers" and "Reconnaissance" attack types were removed from training data but retained in the test set. Figure 1 illustrates the distribution of attack categories across different data subsets. The training data excludes "Fuzzers" and "Reconnaissance," while the test data includes all attack types. The zero-day evaluation subset contains only "Fuzzers," "Reconnaissance," and normal traffic, providing a focused dataset for evaluating the model's ability to detect previously unseen attack patterns. Fig. 1 show the distribution of attack categories across different data subsets.

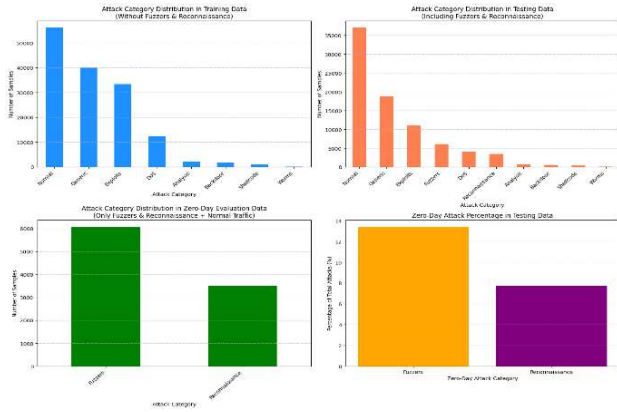


Fig. 1. Distribution of attack categories across different data subsets.

B. Data Preprocessing

The preprocessing pipeline was designed to facilitate robust zero-day attack evaluation. "Fuzzers" and "Reconnaissance" attack categories were deliberately excluded from the training data but retained in the testing sets. A specialized zero-day dataset containing 46,558 samples, composed of unseen attack types and normal traffic, was constructed to evaluate the model's ability to detect novel threats. Data cleaning involved mean imputation for missing values and removal of duplicate records, resulting in 175,341 clean training entries. Categorical features were converted into numerical format using label encoding, with unified mapping across all datasets to ensure consistency.

To handle class imbalance, the SMOTE+Tomek method was applied, producing a balanced dataset with a 1.0000 ratio between attack and normal classes. The original distribution of 90,666 attacks and 56,000 normal instances was restructured into 86,709 samples per class. This method not only generates synthetic data but also removes noisy borderline samples that may hinder learning. StandardScaler normalization was then employed to standardize all numeric features, ensuring equal contribution during training and preventing scale dominance.

Dimensionality reduction started with correlation analysis by eliminating features with correlation coefficients greater than 0.9. Subsequently, SelectKBest with the `f_classif` scoring function was used to identify the top 20 most statistically

significant features for attack classification. These steps reduced multicollinearity and computational load while preserving predictive performance. All preprocessing components were serialized to guarantee consistent data transformation during model deployment. Table II shows the feature selection results with SelectKBest.

TABLE II. FEATURE SELECTION RESULTS WITH SELECTKBEST

proto	sttl	djit	dmean
service	sload	swin	ct_srv_src
state	dload	stcpb	ct_state_ttl
dbytes	dinpkt	dtepb	ct_src_ltm
rate	sjit	smean	is_sm_ips_ports

C. Model Development

This study uses two reinforcement learning (RL) algorithms, Q-Learning and Deep Q-Learning (DQN), to create a system that can adapt to find anomalies. RL lets an agent learn the best ways to make decisions by interacting with an environment in a trial-and-error way to get the most cumulative rewards. We chose Q-Learning because it is simple and works quickly. DQN builds on Q-Learning by adding deep learning to deal with the complicated, high-dimensional input spaces that are common in network traffic data.

Q-Learning is a model-free reinforcement learning (RL) algorithm. It learns the expected utility of actions in a given state using a table called the Q-table. The environment is modeled as a Markov Decision Process (MDP), and the agent updates the Q-values based on the rewards it receives. The update is performed using the following equation:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[r + \gamma \max_{a'} Q(s', a') - Q(s, a) \right] \quad (1)$$

where α is the learning rate, γ is the discount factor, r is the reward, and (s, a, s', a') stand for the current state, action, next state, and next action. Q-Learning can learn the attack from traffic features like the protocol, port, time, and data length. But the limitation of the Q-table will make its performance deteriorate in scenarios where the state space is big or continuous [4].

Deep Q-Learning (DQN) employs a deep neural network instead of the Q-table to approximate the Q-function. This enables the model to generalize across high-dimensional state representations [14]. Neural models are also used in newer research, such as the use of multitask deep learning to identify DDoS attacks in IoT networks, to recognize complex attack patterns in network traffic [15]. DQN uses experience replay through storing past experiences in a memory buffer and sampling them at random during training. This separates the samples and stabilizes the system. For stabilizing learning and avoiding it from deviating in many different directions, there is also a target network that is the same as the primary network employed and updated at intervals [4]. The mathematical representation of DQN is given as follows:

$$Q(s, a; \theta) \approx Q^*(s, a) \quad (2)$$

where $Q(s, a; \theta)$ is the approximated Q-function with parameters θ , and $Q^*(s, a)$ is the optimal Q-function. The network is trained by minimizing the loss function:

$$L(\theta) = E(s, a, r, s') \sim D \left[\left(r + \gamma \max_{a'} Q(s', a'; \theta) - Q(s, a; \theta) \right)^2 \right] \quad (3)$$

where D is the experience replay buffer, γ is the discount factor, and θ represents the parameters of the target network [4]. The DQN model comprises an input layer, two hidden layers having ReLU as the activation function, and an output layer offering Q-values for every possible action [14]. Using ReLU helps with the vanishing gradient problem and increases the rate of convergence. DQN can learn high-level features and learn abstract attack patterns for network anomaly detection. That is, it can detect both known and unknown attacks [4]. It works excellent for dynamic and intricate intrusion detection assignments because it is strong with very few false positives.

D. Model Validation

Model validation is required in order for the anomaly detection model to function optimally and be capable of recognizing patterns of attack. The comparison of output from the model with real conditions via a confusion matrix is how this research quantifies the model [16]. There are four main sections of this matrix: True Positive (TP) for correctly identified anomalies, True Negative (TN) for correctly identified normal data, False Positive (FP) for normal data incorrectly detected as an anomaly, and False Negative (FN) for anomalies that go undetected. Additionally, ROC curves are used to show the trade-off between detection rate and false alarms at different thresholds. The Area Under the Curve (AUC) measures how well the model can distinguish between normal and anomalous traffic.

The validation process is performed on both the training and testing datasets. Evaluating the training data aims to check whether the model is overfitting, while testing assesses the model's generalization ability to unseen data. To measure the model's capability in detecting new types of attacks, a specific subset of the testing data containing zero-day attacks is also used. This subset consists of attack categories that were not involved in the training, thus reflecting the model's ability to detect previously unknown threats. Several evaluation metrics are used including accuracy (overall rate of correct predictions), precision (proportion of true positive predictions), recall (proportion of actual positives correctly identified), and F1 Score (harmonic mean of precision and recall) as shown in Table III and Table IV.

TABLE III. CONFUSION MATRIX

	Predict Normal	Predict Anomaly
Actual Normal	True Negative (TN)	False Positive (FP)
Actual Anomaly	False Negative (FN)	True Positive (TP)

TABLE IV. PERFORMANCE METRICS

Matrix	Description	Equation
Accuracy	Overall rate of correct predictions	$\frac{TP + TN}{TP + FP + TN + FN}$
Precision	Proportion of true positive predictions	$\frac{TP}{TP + FP}$
Recall	Proportion of actual positives that were correctly identified	$\frac{TP}{TP + FN}$
F1 Score	Harmonic mean of precision and recall	$2 \times \frac{(Precision \times Recall)}{(Precision + Recall)}$

IV. RESULTS AND DISCUSSION

In reinforcement learning (RL), agents interact with their environment to maximize cumulative rewards and learn effective decision-making strategies [17]. In this study, we develop an anomaly detection model for network traffic using two RL algorithms: Q-Learning and Deep Q-Learning (DQN), chosen for their capability to learn optimal policies and adapt to dynamic conditions. To ensure transparency and reproducibility, the full implementation of our approach is publicly available on GitHub (<https://github.com/adhitamaw/Anomaly-Detection-using-Reinforcement-Learning>). This resource enables researchers and practitioners to replicate our results or extend the proposed methodology.

A. Experimental Preparation

This research evaluates the effectiveness of Reinforcement Learning (RL) algorithms, namely Q-Learning and Deep Q-Learning (DQN), in detecting network anomalies, particularly zero-day attacks. The dataset used is the modified UNSW-NB15, from which two types of attacks (Fuzzers and Reconnaissance) were removed from the training data and only included in the testing data. Additionally, a specific test dataset was created that only includes Fuzzers and Reconnaissance attacks. The aim is to simulate attack scenarios that the model has never encountered before.

The preprocessing steps include handling missing data, encoding categorical features, dimensionality reduction, and normalization. Feature selection using SelectKBest with ANOVA F-value identified the 20 most informative features for classification. To address class imbalance, we applied SMOTE+Tomek resampling, which resulted in 86,709 samples per class (173,418 total). For computational efficiency, our experiments utilized 50% of this resampled data (86,709 samples total).

TABLE V. DISTRIBUTION OF DATASET IN THE EXPERIMENT

Dataset	Sample Size	Normal (%)	Attack (%)	Attack Categories
Training	86,709 (50%)	50.0%	50.0%	Analysis, Backdoor, DoS, Exploits, Generic, Shellcode, Worms
General Testing	82,332	56.0%	44.0%	All categories
Zero-day Testing	46,558	62.5%	37.5%	Only Fuzzers & Reconnaissance + Normal

This distribution ensures that the model is trained on known attack data and then tested on data that includes both known and previously untrained attacks, thereby measuring the model's generalization ability and detection capability for new (zero-day) attacks.

B. Hyperparameter Tuning

To optimize the performance of the Q-Learning and DQN models, hyperparameter tuning was conducted through five different schemes. The hyperparameter values were selected based on empirical evidence to ensure optimal model performance, especially in detecting zero-day attacks. Table VI

present the Q-Learning hyperparameter tuning. Meanwhile, Table V show the DQN hyperparameter tuning.

TABLE VI. Q-LEARNING HYPERPARAMETER TUNING

Scheme	Learning Rate (α)	Discount Factor (γ)	Initial Epsilon (ϵ)	Epsilon Decay	Minimum Epsilon
1	0.01	0.8	1.0	0.8	0.05
2	0.1	0.95	1.0	0.95	0.2
3	0.03	0.99	0.8	0.99	0.3
4	0.5	0.9	0.7	0.9	0.1
5	0.2	0.7	0.9	0.85	0.01

TABLE VII. DQN HYPERPARAMETER TUNING

Scheme	Learning Rate	Epsilon Decay	Batch Size	Buffer Size
1	0.0001	0.93	32	50000
2	0.0003	0.95	64	50000
3	0.0005	0.97	128	50000
4	0.001	0.98	256	50000
5	0.003	0.99	512	50000

Both models used 30 states to represent network traffic patterns and a binary action space (2 actions) to classify traffic as normal or anomalous. Additionally, 42 random states were used to ensure reproducibility. The training process was conducted for 30 episodes for Q-Learning, while some DQN parameters remained constant across all schemes, such as a discount factor (γ) of 0.95, initial exploration rate (ϵ) of 1.0, minimum exploration rate of 0.15, target network update frequency every 3 episodes, with a total of 15 training episodes. The hyperparameter tuning process was carried out systematically to identify the optimal configuration that could maximize detection accuracy, with a particular focus on the ability to detect zero-day attacks.

C. Model Validation Results

The performance evaluation of Q-Learning and Deep Q-Learning (DQN) on standard datasets and zero-day attacks shows significant differences between approaches. Models were evaluated using the same test dataset to ensure fair comparison. Q-Learning showed competitive performance, with Scheme 1 achieving the best results (accuracy 94.18%, F1 0.9447) using a small learning rate (0.01), high discount factor (0.8), and good exploration control (epsilon decay 0.8) as shown in Table VIII.

TABLE VIII. Q-LEARNING PERFORMANCE ON STANDARD TEST DATA

Scheme	Accuracy	Precision	Recall	F1 Score	FPR	AUC
1	0.9418	0.9903	0.9030	0.9447	0.0108	0.9825
2	0.9259	0.9950	0.8699	0.9282	0.0054	0.9690
3	0.9235	0.9943	0.8661	0.9257	0.0061	0.9727
4	0.9181	0.9760	0.8727	0.9215	0.0262	0.9577
5	0.9235	0.9788	0.8801	0.9268	0.0233	0.9666

Meanwhile, DQN demonstrated significantly superior performance, with Scheme 1 achieving the highest accuracy (99.09%) and Scheme 5 achieving the highest F1 score (0.9972). Scheme 1 uses a small learning rate (0.0001), moderate epsilon decay (0.93), and a small batch size (32) as shown in Table IX.

TABLE IX. DQN PERFORMANCE ON STANDARD TEST DATA

Scheme	Accuracy	Precision	Recall	F1	FPR	AUC
1	0.9909	0.9840	0.9998	0.9918	0.0199	0.9999
2	0.9888	0.9941	0.9855	0.9898	0.0071	0.9898
3	0.9958	0.9971	0.9954	0.9962	0.0036	0.9973
4	0.9945	0.9935	0.9965	0.9950	0.0079	0.9991
5	0.9970	0.9969	0.9975	0.9972	0.0038	0.9994

DQN Scheme 1 identified most attacks with 99.98% recall, though with a slightly higher false positive rate of 1.99%. Compared to Q-Learning Scheme 1, DQN showed an accuracy improvement of 4.91%, precision decrease of 0.63%, and recall improvement of 9.68%. The AUC value of 0.9999 for DQN confirms its superior discrimination ability. After evaluating all schemes, we identified the best performing configuration for each algorithm and compared their effectiveness on zero-day attack detection. Table X shows the comparison of the best Q-Learning and DQN schemes on zero-day attack data. While, Table XI presents the confusion matrix.

TABLE X. COMPARISON OF BEST Q-LEARNING AND DQN SCHEMES ON ZERO-DAY ATTACK DATA

Model	Accuracy	Precision	Recall	F1 Score	FPR	AUC
Q-Learning	0.9187	0.9393	0.6456	0.7653	0.0108	0.9472
DQN	0.9840	0.9283	0.9991	0.9624	0.0199	0.9994

TABLE XI. CONFUSION MATRICES FOR BEST Q-LEARNING AND DQN SCHEMES ON ZERO-DAY ATTACK DATA

Model	TP	FP	TN	FN
Q-Learning	6171	399	36601	3387
DQN	9549	737	36263	9

The comparative analysis reveals a substantial performance gap between the two approaches. While Q-Learning Scheme 1 achieved a moderate detection rate of 64.56% for zero-day attacks, DQN Scheme 1 demonstrated exceptional detection capabilities with a 99.91% detection rate. The confusion matrices illustrate this stark difference clearly - DQN misclassified only 9 attack instances as normal, while Q-Learning misclassified 3,387 attacks. Although Q-Learning shows slightly fewer false positives (399 versus DQN's 737), its overall ability to detect zero-day attacks is significantly weaker. DQN showed significant advantages: a 35.35% higher detection rate, a slight precision decrease of 1.10%, and a 19.71% higher F1 score.

ROC curve analysis revealed significant performance differences between the models. DQN Scheme 1 demonstrated nearly perfect classification with AUC values of 0.9999 for standard test data and 0.9994 for zero-day attacks, showing minimal degradation (0.0005) when facing novel threats. In contrast, Q-Learning exhibited a more substantial performance drop, with AUC decreasing from 0.9825 on standard data to 0.9472 on zero-day scenarios. Fig. 2 shows the ROC curves of both comparison methods.

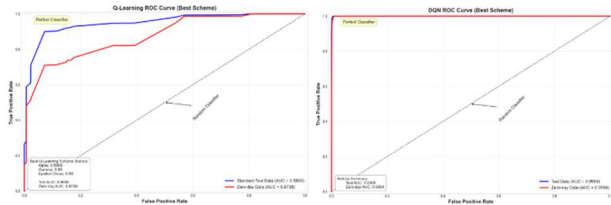


Fig. 2. ROC Curves for Best Q-Learning and DQN Schemes on Standard and Zero-Day Data.

TABLE XII. DETECTION RATE DECREASE FROM STANDARD TO ZERO-DAY

Model	Scheme 1	Scheme 2	Scheme 3	Scheme 4	Scheme 5
DQN	0.07%	5.43%	1.74%	1.30%	0.87%
Q-Learning	25.74%	34.75%	36.45%	34.31%	31.21%

All DQN schemes showed outstanding generalization capability with minimal performance degradation (<6% for all schemes). In contrast, Q-Learning showed much weaker generalization, with performance drops of 25-36% when facing zero-day attacks as shown in Table XII.

TABLE XIII. PERFORMANCE COMPARISON OF OTHER MODELS ON STANDARD TEST DATA

Metric	Q-Learning	DQN	Random Forest	SVM
Accuracy	0.9418	0.9909	0.8978	0.8740
Precision	0.9903	0.9840	0.9462	0.9065
Recall/DR	0.9030	0.9998	0.8634	0.8598
F1 Score	0.9447	0.9918	0.9029	0.8826
FPR	0.0108	0.0199	0.0601	0.1086
AUC	0.9825	0.9999	0.9667	0.9337

TABLE XIV. PERFORMANCE COMPARISON OF OTHER MODELS ON ZERO-DAY DATA

Metric	Q-Learning	DQN	Random Forest	SVM
Accuracy	0.9187	0.9840	0.8360	0.8063
Precision	0.9393	0.9283	0.6508	0.5314
Recall/DR	0.6456	0.9991	0.4337	0.4768
F1 Score	0.7653	0.9624	0.5205	0.5026
FPR	0.0108	0.0199	0.0601	0.1086
AUC	0.9472	0.9994	0.8605	0.7588

Based on the SHAP analysis in Fig. 3 and Fig. 4, we identified key features influencing model predictions. Figure 3 shows feature importance comparison between regular and zero-day attacks, where feature 4 (dbytes) dominates for regular attacks while feature 2 (service) becomes critical for zero-day detection. This shift highlights how different traffic characteristics matter when facing novel threats. Fig. 4 confirms features 2 (service), 4 (dbytes), 7 (sload), 17 (ct_srv_src), and 12 (swin) have the strongest impact. The service feature particularly stands out for zero-day detection, suggesting network services provide crucial indicators for distinguishing malicious traffic. Similarly, dbytes shows substantial influence for regular attacks but varies in zero-day scenarios. The sload feature demonstrates significant impact, aligning with cybersecurity knowledge where unusual data rates often indicate attacks. These findings suggest monitoring specific traffic characteristics enhances

intrusion detection capabilities, confirming our feature selection effectiveness for previously unseen attack vectors.

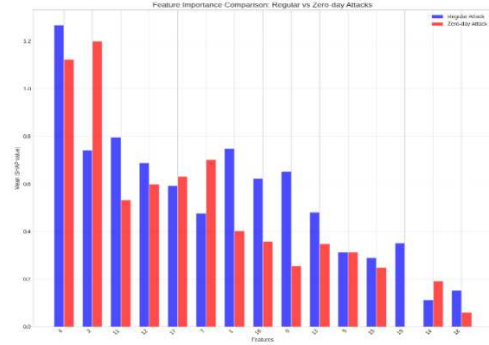


Fig. 3. Feature Importance Comparison.

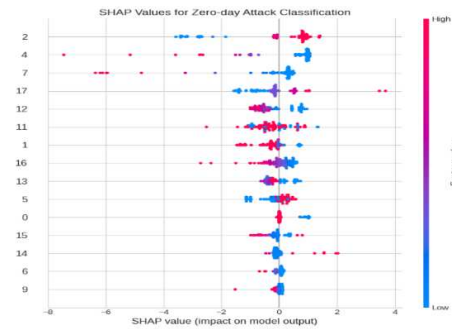


Fig. 4. SHAP Values for Zero-day Classification.

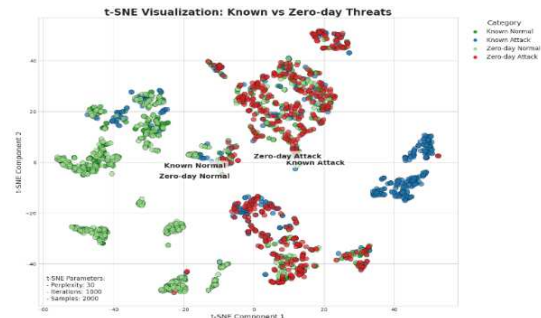


Fig. 5. t-SNE Visualization.

The t-SNE visualization in Fig. 5 provides insight into DQN's superior zero-day detection by revealing how the model organizes traffic types in feature space. The visualization shows four distinct clusters (Known Normal, Known Attack, Zero-day Normal, and Zero-day Attack), with DQN creating clear boundaries between normal and attack traffic regardless of training exposure. This indicates strong generalization capabilities. Zero-day attack clusters are positioned between known attacks and normal traffic, suggesting the model learned a generalized concept of "attackness" beyond specific signatures. While known attacks form dense clusters, zero-day attacks display more distributed patterns partially overlapping with known attack regions. This demonstrates how DQN's architecture transforms raw feature space into a representation where attack-related features are enhanced and irrelevant ones suppressed. As a result, the model learns underlying anomalous behavior patterns rather than memorizing specific signatures,

successfully classifying previously unseen traffic patterns and creating robust decision boundaries that generalize well to new threats.

TABLE XV. COMPUTATIONAL EFFICIENCY COMPARISON

Model	Training Time (sec)	Memory Usage (MB)	Model Size (MB)	Inference Speed (samples/sec)
Q-Learning	36.03	1,247.39	0.000458	997,931
DQN	4,852.5	2,572.1	0.03	3,623.4

DQN in Table XV achieved superior detection performance, efficiency considerations reveal important trade-offs. As shown in our computational comparison, Q-Learning demonstrated major advantages by requiring only 36.03 seconds of training time (0.74% of DQN's 4,852.5 seconds), 1,247.39 MB of memory (48.5% of DQN's 2,572.1 MB), and a dramatically smaller model size (0.000458 MB vs. 0.03 MB). It also achieved approximately 275 times faster inference speed, processing 997,931 samples per second compared to DQN's 3,623.4. Despite DQN's exceptional zero-day detection capabilities, these high computational demands limit its deployment in resource-constrained or real-time environments. To address this, we recommend a hybrid approach that uses lightweight Q-Learning models for initial threat screening and deploys DQN on dedicated hardware for deeper analysis. Organizations may also consider offline pre-training, scheduled model updates during low-traffic periods, and model compression to align performance with available resources and security needs.

V. CONCLUSION

This study evaluated Q-Learning and Deep Q-Learning (DQN) for network intrusion detection with a focus on zero-day attacks. DQN significantly outperformed Q-Learning across all key metrics, with DQN Scheme 1 achieving 99.09% accuracy and a 99.98% detection rate on standard test data compared to Q-Learning Scheme 1's 94.18% accuracy and 90.30% detection rate. Most notably, DQN demonstrated exceptional zero-day attack detection, sustaining a 99.91% detection rate with minimal degradation (<1%), whereas Q-Learning dropped to 64.56%. Despite this superior detection performance, Q-Learning offers considerable computational efficiency: short training times, small model sizes, and rapid inference, making it more suitable for resource-constrained environments. In practical cybersecurity operations, these findings imply that DQN-based solutions are well-suited for enterprise-scale systems, cloud infrastructures, data center networks, and other high-security environments emphasizing robust detection over resource efficiency. Q-Learning provides a viable option for organizations such as branch offices, small or embedded IoT networks, and industrial control systems operating under strict hardware or latency constraints. By tailoring the RL approach to the particular operational requirements (whether real-time detection, minimal resource utilization, or the capacity to handle evolving threats), this study's results demonstrate a clear pathway for integrating adaptive anomaly detection into heterogeneous cybersecurity deployments.

REFERENCES

- [1] A. Gueriani, H. Kheddar, and A. C. Mazari, "Deep reinforcement learning for intrusion detection in IoT: A survey," in *Proc. 2023 2nd Int. Conf. Electron., Energy Meas. (IC2EM)*, Nov. 2023, vol. 1, pp. 1–7, doi: 10.1109/IC2EM12345.2023.00001.
- [2] M. M. Al-Nawashi, O. M. Al-Hazaimah, N. M. Tahat, N. Gharaibeh, W. A. Abu-Ain, and T. Abu-Ain, "Deep reinforcement learning-based framework for enhancing cybersecurity," *Int. J. Interact. Mobile Technol.*, vol. 19, no. 3, 2025.
- [3] J. K. Yego, N. Kiget, and D. Samoei, "A deep reinforcement learning approach to modelling an intrusion detection system using asynchronous advantage actor-critic algorithm," *J. Res. Innov. Implic. Educ.*, vol. 6, no. 1, pp. 441–452, 2022.
- [4] H. Alavizadeh and J. Jang-Jaccard, "Deep Q-learning based reinforcement learning approach for network intrusion detection," *Computers*, vol. 11, no. 3, p. 41, 2022, doi: 10.3390/computers11030041.
- [5] M. H. Modirrousta, P. F. Arani, and M. A. Shooreshdeli, "Analysis of anomalous behavior in network systems using deep reinforcement learning with CNN architecture," Nov. 2022. [Online]. Available: <http://arxiv.org/abs/2211.16304>
- [6] A. Tellache, A. Mokhtari, A. A. Korba, and Y. Ghamri-Doudane, "Multi-agent reinforcement learning-based network intrusion detection system," in *Proc. NOMS 2024 IEEE Netw. Oper. Manag. Symp.*, May 2024, pp. 1–9, doi: 10.1109/NOMS2024.1234567.
- [7] N. Sharma, N. S. Yadav, and S. Sharma, "Classification of UNSW-NB15 dataset using exploratory data analysis and ensemble learning," *EAI Endorsed Trans. Ind. Netw. Intell. Syst.*, vol. 8, no. 29, p. e4, 2021.
- [8] M. Ali, S. Pervez, S. E. Hosseini, and M. K. Siddhu, "Evaluation and detection of cyberattack in IoT-based smart city networks using machine learning on the UNSW-NB15 dataset," *Int. J. Online Biomed. Eng. (iJOE)*, vol. 21, no. 2, 2025.
- [9] H. Randika, P. Sukarno, and A. A. Wardana, "Attack detection on edge IIoT using ensemble learning and XAI model transparency," in *Proc. 2025 Int. Conf. Adv. Data Sci., E-Learning Inf. Syst. (ICADEIS)*, Feb. 2025, pp. 1–6, doi: 10.1109/ICADEIS2025.1234567.
- [10] A. Ahmadi, S. S. Sharif, and Y. M. Banad, "A comparative study of sampling methods with cross-validation in the FedHome framework," *IEEE Trans. Parallel Distrib. Syst.*, 2025, doi: 10.1109/TPDS.2025.1234567.
- [11] M. M. Ahsan, M. P. Mahmud, P. K. Saha, K. D. Gupta, and Z. Siddique, "Effect of data scaling methods on machine learning algorithms and model performance," *Technologies (Basel)*, vol. 9, no. 3, p. 52, 2021, doi: 10.3390/technologies9030052.
- [12] C. A. Rickert, M. Henkel, and O. Lieleg, "An efficiency-driven, correlation-based feature elimination strategy for small datasets," *APL Mach. Learn.*, vol. 1, no. 1, 2023, doi: 10.1063/5.0134567.
- [13] H. Abouloifa and M. Bahaj, "Predicting late delivery in supply chain 4.0 using feature selection: A machine learning model," in *Proc. 2022 5th Int. Conf. Adv. Commun. Technol. Netw. (CommNet)*, Dec. 2022, pp. 1–5, doi: 10.1109/CommNet2022.1234567.
- [14] V. Mnih et al., "Human-level control through deep reinforcement learning," *Nature*, vol. 518, no. 7540, pp. 529–533, 2015, doi: 10.1038/nature14236.
- [15] M. F. Abyandani, P. Sukarno, and A. A. Wardana, "Distributed denial-of-service (DDoS) detection using multitask learning based on deep learning," in *Proc. 2025 Int. Conf. Artif. Intell. Inf. Commun. (ICAIIIC)*, Feb. 2025, pp. 383–388, doi: 10.1109/ICAIIIC2025.1234567.
- [16] G. Y. Kim, S. M. Lim, and I. C. Euom, "A study on performance metrics for anomaly detection based on industrial control system operation data," *Electronics (Basel)*, vol. 11, no. 8, p. 1213, 2022, doi: 10.3390/electronics11081213.
- [17] M. Baich, T. Hamim, N. Sael, and Y. Chemlal, "Machine learning for IoT-based networks intrusion detection: A comparative study," *Procedia Comput. Sci.*, vol. 215, pp. 742–751, 2022, doi: 10.1016/j.procs.2022.12.123.